# TP, FP, FN, TN: the one sheet

Every metric is a ratio of cells of one $2 \times 2$ table. Decide which cells you know, and everything else follows.

## 1. The table, and how to read the four letters

| | **alert** (predict +) | **no alert** (predict −) | row sum |
|---|---|---|---|
| **truth: attack** (+) | $TP$ | $FN$ (missed attack) | attacks $= TP + FN$ |
| **truth: benign** (−) | $FP$ (false alarm) | $TN$ | benign $= FP + TN$ |
| column sum | alerts $= TP + FP$ | silent $= FN + TN$ | $N$ |

**Second letter = what the model said** (P = it alerted, N = it stayed silent). **First letter = was the model right?** (T = yes, F = no). So $FN$ = "the model said *negative*, and that was *false*": a missed attack. The two red cells are the two kinds of mistake, and every cost-based decision is a trade between them. **Rows are the truth, columns are the prediction.** Each metric divides a cell by its row total or its column total; that choice is the whole meaning of the metric.

## 2. The metrics: formula, denominator, question, and what moves them

| metric (aliases) | formula | divides by | the question it answers | prevalence-dependent? | trivial rules give |
|---|---|---|---|---|---|
| **recall** (sensitivity, TPR) | $\frac{TP}{TP+FN}$ | attack *row* | Of the real attacks, what fraction did we catch? | no | all-silent: 0; all-alert: 1 |
| **miss rate** (FNR) | $\frac{FN}{TP+FN}$ $= 1 - \text{recall}$ | attack row | Of the real attacks, what fraction slipped through? | no | |
| **false-positive rate** (FPR) | $\frac{FP}{FP+TN}$ | benign *row* | Of the benign flows, what fraction did we wrongly alert on? | no | all-silent: 0; all-alert: 1 |
| **specificity** (TNR) | $\frac{TN}{TN+FP}$ $= 1 - \text{FPR}$ | benign row | Of the benign flows, what fraction did we leave alone? | no | |
| **precision** (PPV) | $\frac{TP}{TP+FP}$ | alert *column* | Of our alerts, what fraction were real attacks? | **yes** | all-silent: undefined (0/0); all-alert: = prevalence |
| **false discovery rate** (FDR) | $\frac{FP}{TP+FP}$ $= 1 - \text{prec.}$ | alert column | Of our alerts, what fraction were noise? | **yes** | |
| **accuracy** | $\frac{TP+TN}{N}$ | everything | What fraction of all decisions were right? | **yes** | all-silent: $1 - \pi$ |
| **balanced accuracy** | $\frac{1}{2}(\text{recall} + \text{spec.})$ | one row each | Average of the two per-class hit rates | no | any constant rule: 0.5 |
| $F_1$ | $\frac{2\,PR}{P+R}$ | (precision, recall) | Are precision and recall *both* high? (harmonic mean: the smaller dominates) | **yes** (via $P$) | all-alert: $\approx 2\pi$ |
| **prevalence** $\pi$ | $\frac{TP+FN}{N}$ | everything | What fraction of flows are attacks? | is the prevalence | |
| **alert rate** | $\frac{TP+FP}{N}$ | everything | What fraction of flows do we alert on? (analyst load) | yes | |

**Do not confuse**

**Row metrics versus column metrics.** Recall and FPR divide by a *truth row*, so they describe the classifier on one class and do not care how many flows of the other class exist. Precision divides by the *alert column*, which mixes both classes, so it changes when the prevalence changes even if the classifier does not. The link is Bayes' rule:

$$\text{precision} = \frac{\text{recall} \cdot \pi}{\text{recall} \cdot \pi + \text{FPR} \cdot (1 - \pi)}.$$

Recall 0.7 and FPR 0.018 give precision 0.28 at $\pi = 1\%$, 0.81 at 10%, and 0.04 at 0.1%. Same model, three precisions.

## 3. Which cells do you actually know? (the step people skip)

Before computing anything, mark each of the four cells as *observed* or *unknown*. A metric is computable only if every cell in its formula is observed.

| to compute | you need |
|---|---|
| precision, FDR, alert rate | the alert column: $TP$, $FP$ |
| recall, miss rate | the attack row: $TP$ *and* $FN$ |
| FPR, specificity | the benign row: $FP$ *and* $TN$ |
| accuracy, $F_1$, balanced acc., prevalence | all four cells |

**Who sees which cells?** Analysts investigating alerts see the alert column only. A labelled test set shows all four. A random audit of the *silent* column estimates $FN$ and $TN$, which unlocks everything.

**Worked numbers: the "alerts only" situation**

100,000 flows/day, 2,000 alerts, 300 confirmed. Nobody looks at the 98,000 silent flows.

| | alert | silent |
|---|---|---|
| attack | $TP = 300$ | $FN = ?$ |
| benign | $FP = 1700$ | $TN = ?$ |

**Computable:** precision $= 300/2000 = 0.15$; FDR 0.85; alert rate 2%.
**Not computable:** recall (needs $FN$), FPR (needs $TN$), accuracy, prevalence.
**After auditing 1,000 silent flows and finding 4 attacks:** $FN \approx 0.004 \times 98{,}000 = 392$, $TN \approx 97{,}608$. Then recall $\approx 300/692 = 0.43$, FPR $\approx 1700/99{,}308 = 0.017$, prevalence $\approx 0.7\%$, accuracy $\approx 0.979$. All of it rests on 4 events, so recall is really "somewhere between 0.3 and 1".

## 4. A six-step routine for any such situation

1. **Name the positive class.** Positive = the rare thing you are hunting (attack, fraud, disease). Every letter below refers to it.
2. **Translate each sentence into a cell or a sum.** "alerted on 2,000" $\rightarrow TP + FP = 2000$. "300 confirmed" $\rightarrow TP = 300$. "100 attacks in the test set" $\rightarrow TP + FN = 100$. "nobody looked at the rest" $\rightarrow FN$, $TN$ unknown.
3. **Fill the table**, writing "?" in unknown cells. Use row and column sums to solve for missing cells when the totals are given.
4. **Compute only what the known cells allow.** Say explicitly which metrics are not computable and what information would unlock them.
5. **Compare with the trivial rules.** All-silent has accuracy $1 - \pi$ and recall 0; all-alert has recall 1 and precision $\pi$. A model must beat the relevant one on the metric that matters.
6. **Ask what changes if the prevalence changes.** Row metrics (recall, FPR) stay; column metrics (precision, accuracy, $F_1$) move. If deployment prevalence differs from the test set, recompute precision with the Bayes formula above.

## 5. Worked example with all four cells (Tutorial 1, Problem 4c)

Test set of 10,000 flows with 100 attacks; $TP = 70$, $FP = 180$. Row sums give $FN = 100 - 70 = 30$ and $TN = 9{,}900 - 180 = 9{,}720$.

| | | |
|---|---|---|
| recall $= 70/100 = 0.70$ | precision $= 70/250 = 0.28$ | FPR $= 180/9900 = 0.018$ |
| accuracy $= 9790/10000 = 0.979$ | $F_1 = 2(0.28)(0.70)/0.98 = 0.40$ | all-silent accuracy $= 9900/10000 = 0.99$ |

Reading: we catch 70% of attacks; 72% of our alerts are noise; we bother 1.8% of benign flows. All-silent scores accuracy 0.99, *higher* than the model, with recall 0. That single comparison is why accuracy is never the headline metric with a rare positive class.

## 6. Five traps

1. **Swapping precision and recall.** Precision has $FP$ in the denominator (it is about the *alerts*); recall has $FN$ (it is about the *attacks*). "Precise" = few false alarms; "recalls everything" = few misses.
2. **"1% false-positive rate" ≠ "1% of alerts are false".** FPR is $FP$ over *benign flows*; the fraction of alerts that are false is the FDR, 1− precision. With rare positives, a 1% FPR can mean 90% of alerts are false.
3. **Accuracy on imbalanced data.** It is $1 - \pi$ for doing nothing. Report recall and precision (or FPR) instead, and the counts.
4. **Moving the threshold.** Lowering it can only raise recall and raise FPR; precision can go either way. There is no threshold that improves everything.
5. **Metrics on a rare class are noisy.** Recall is an average over the attacks only: 70 of 100 gives standard error $\sqrt{0.7 \times 0.3/100} \approx 0.05$, so "0.70" means "0.60 to 0.80". Precision on 250 alerts is similarly loose. Always ask for the counts behind a rate.